

Revisiting the DimmerSwitch Class

In the following examples, we'll continue with the `DimmerSwitch` class from Chapter 2 (Listing 2-5). The `DimmerSwitch` class already has two instance variables: `self.isOn` and `self.brightness`. The only modification we'll make is to add a `self.label` instance variable so each object we create can be identified easily in the output when we run our program. These variables are created and assigned initial values in the `__init__()` method. They are then accessed or modified in the five other methods of the class.

[*Listing 3-1*](#) provides some test code to create three `DimmerSwitch` objects from the `DimmerSwitch` class, which we'll use in our mental models. I'll call various methods for each of the `DimmerSwitch` objects.

File: OO_DimmerSwitch_Model1.py

```
# Create first DimmerSwitch, turn it on, and raise the level twice
oDimmer1 = DimmerSwitch('Dimmer1')
oDimmer1.turnOn()
oDimmer1.raiseLevel()
oDimmer1.raiseLevel()

# Create second DimmerSwitch, turn it on, and raise the level 3 times
oDimmer2 = DimmerSwitch('Dimmer2')
oDimmer2.turnOn()
oDimmer2.raiseLevel()
oDimmer2.raiseLevel()
oDimmer2.raiseLevel()

# Create third DimmerSwitch, using the default settings
oDimmer3 = DimmerSwitch('Dimmer3')

# Ask each switch to show itself
oDimmer1.show()
oDimmer2.show()
oDimmer3.show()
```

Listing 3-1: Creating three `DimmerSwitch` objects and calling various methods on each

When run with our `DimmerSwitch` class, this code gives the following output: